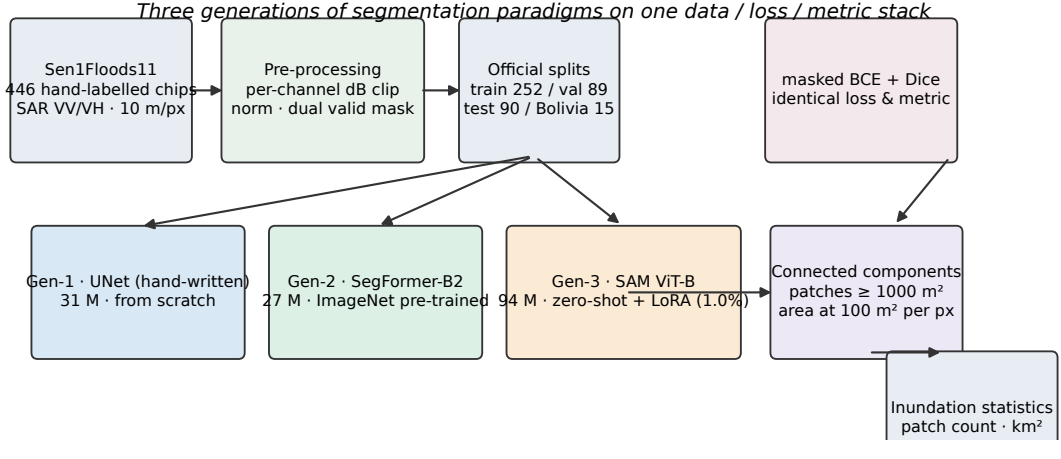
Three generations of segmentation paradigms on one data / loss / metric stack
Sen1Floods11
446 hand-labelled chips
SAR VV/VH · 10 m/px
Pre-processing
per-channel dB clip
norm · dual valid mask
Official splits
train 252 / val 89
test 90 / Bolivia 15
masked BCE + Dice
identical loss & metric
Gen-1 · UNet (hand-written)
31 M · from scratch
Gen-2 · SegFormer-B2
27 M · ImageNet pre-trained
Gen-3 · SAM ViT-B
94 M · zero-shot + LoRA (1.0%)
Connected components
patches ≥ 1000 m²
area at 100 m² per px
Inundation statistics
patch count · km²